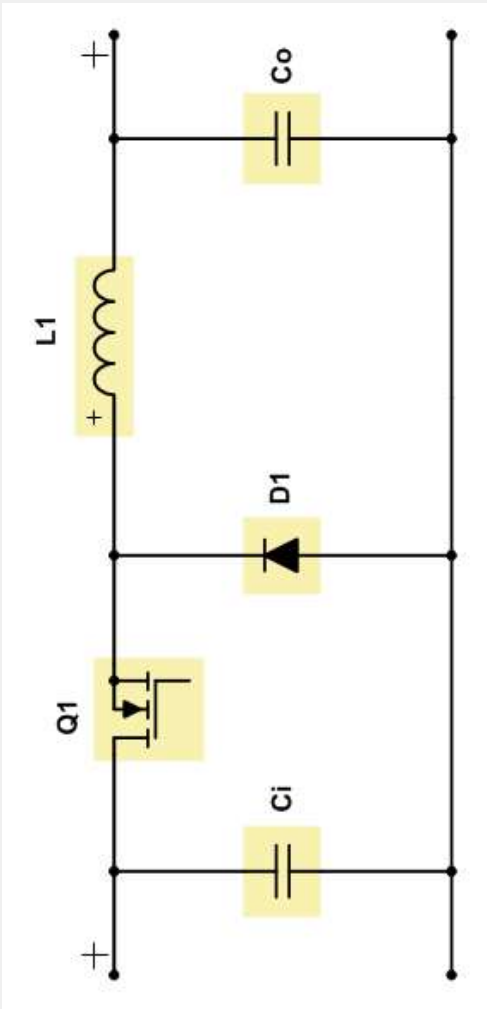


Design Values

Minimum Input Voltage:	36	V
Maximum Input Voltage:	60	V
Output Voltage:	24	V
Output Current:	3	A
Switching Frequency:	500	kHz
Diode Voltage Drop:	0.7	V
Inductor Current Ripple:	15	%



Recommended Value

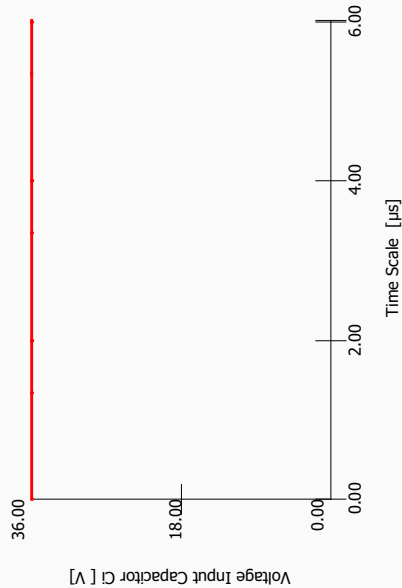
Inductanc... **65.11** μH

Choose Value

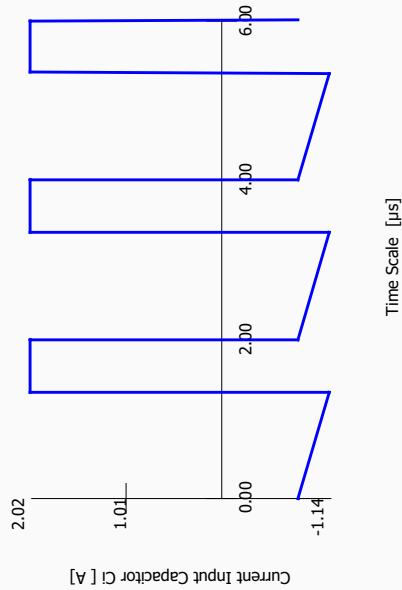
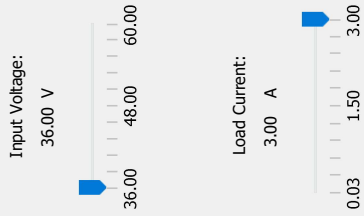
Inductanc... μH

Calculated Values at Input Voltage:				36.00	V	Load Current:	3.00	A
Period:	2.00	μs	Input Power:	72.69	W	Input Current:	2.02	A
Duty Cycle:	67.30	%	Output Power:	72.00	W	Current Ripple:	0.32	A
On-Time:	1.35	μs	Diode Losses:	0.69	W		10.77	%
Off-Time:	0.65	μs						
Zero-Time:	0.00	μs						

1. The minimum inductance is calculated to stay below the specified current ripple over the entire input voltage range.
2. Under the hood of low-voltage DC/DC converters: <https://www.ti.com/lit/slip206>
3. Efficiency of synchronous versus nonsynchronous buck converters: <https://www.ti.com/lit/an/slyt358/slyt358.pdf>
4. Understanding buck power stages in switchmode power supplies: <https://www.ti.com/lit/an/slua057/slua057.pdf>
5. The default values show the LMR14020S Step-Down Converter Evaluation Module LMR14020SEVM: <https://www.ti.com/tool/LMR14020SEVM>
6. The LMR14020 is a SIMPLE SWITCHER, 4-V to 40-V, 2-A 2.2-MHz step-down converter with Low IQ: <https://www.ti.com/product/LMR14020>

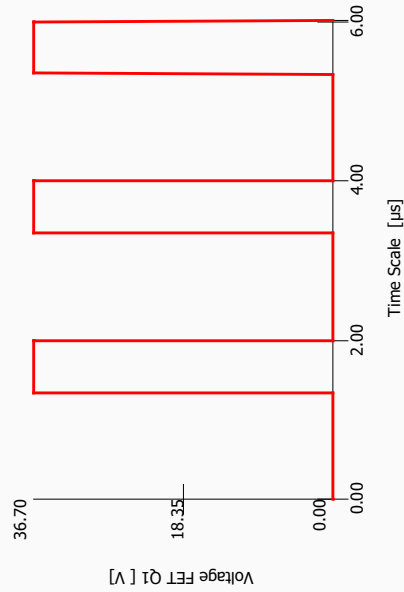


Min. Voltage Input Capacitor C_i : 36.00 V
Max. Voltage Input Capacitor C_i : 36.00 V

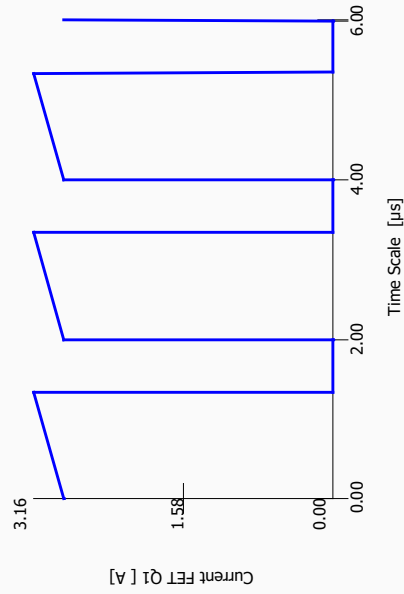
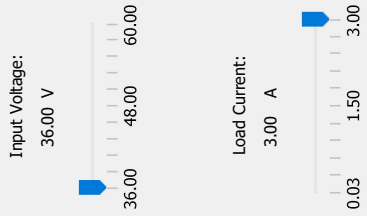


Min. Current Input Capacitor C_i : -1.14 A
Max. Current Input Capacitor C_i : 2.02 A
RMS Current Input Capacitor C_i : 1.41 A
Avg. Current Input Capacitor C_i : 0.00 A
AC Current Input Capacitor C_i : 1.41 A

Close

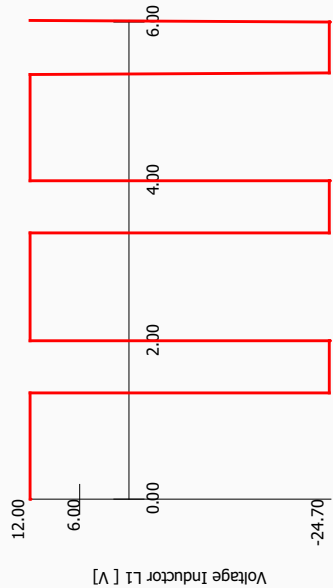


Min. Voltage FET Q1: 0.00 V
Max. Voltage FET Q1: 36.70 V

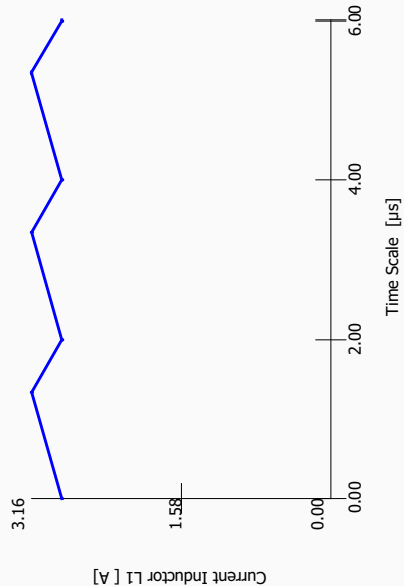
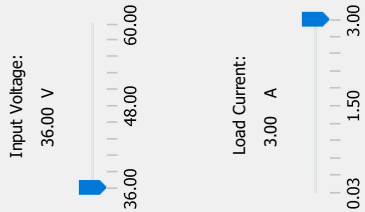


Min. Current FET Q1: 2.84 A
Max. Current FET Q1: 3.16 A
RMS Current FET Q1: 2.46 A
Avg. Current FET Q1: 2.02 A
AC Current FET Q1: 1.41 A

Close

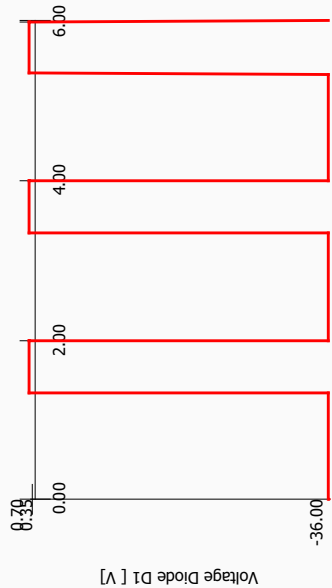


Min. Voltage Inductor L1: -24.70 V
Max. Voltage Inductor L1: 12.00 V

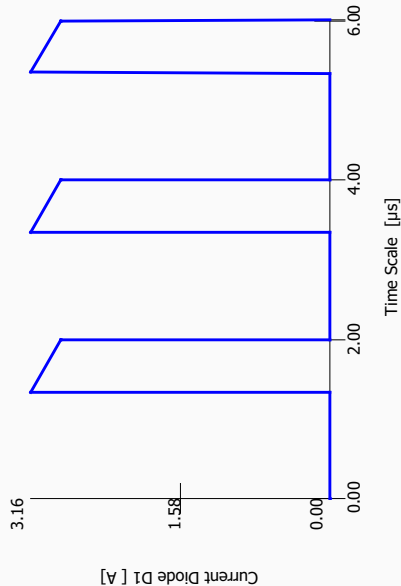
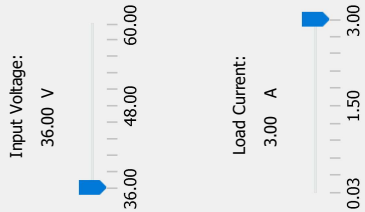


Min. Current Inductor L1: 2.84 A
Max. Current Inductor L1: 3.16 A
RMS Current Inductor L1: 3.00 A
Avg. Current Inductor L1: 3.00 A
AC Current Inductor L1: 0.06 A

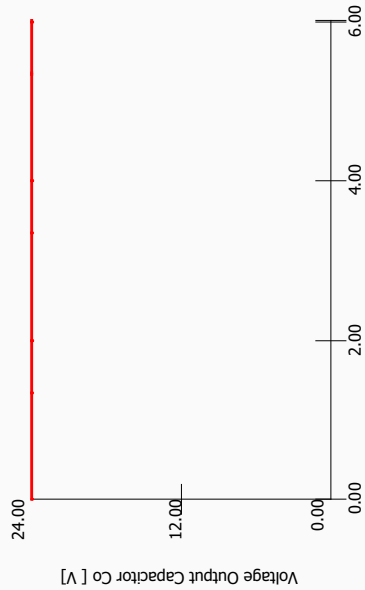
Close



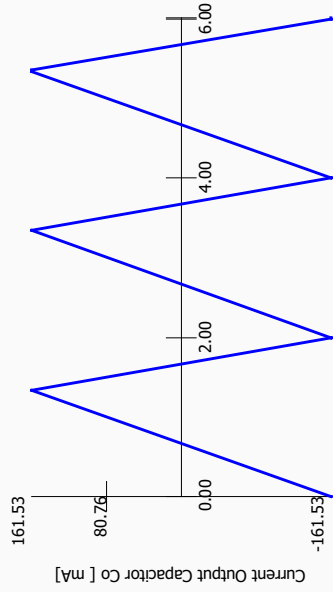
Min. Voltage Diode D1: -36.00 V
Max. Voltage Diode D1: 0.70 V



Min. Current Diode D1: 2.84 A
Max. Current Diode D1: 3.16 A
RMS Current Diode D1: 1.72 A
Avg. Current Diode D1: 0.98 A
AC Current Diode D1: 1.41 A



Min. Voltage Output Capacitor Co: 24.00 V
Max. Voltage Output Capacitor Co: 24.00 V



Time Scale [µs]
Min. Current Output Capacitor Co: -161.53 mA
Max. Current Output Capacitor Co: 161.53 mA
RMS Current Output Capacitor Co: 93.26 mA
Avg. Current Output Capacitor Co: 0.00 mA
AC Current Output Capacitor Co: 93.26 mA

Close